RAJAT BUTOLA (傅能杰)

(+886) 905-048-057 | rajatbutola2@gmail.com | Taipei, Taiwan
LinkedIn | GitHub

PROFESSIONAL SUMMARY

AI Engineer & PhD in EECS with expertise in building and deploying advanced AI/ML systems at the intersection of semiconductor engineering and deep learning. Experienced in developing data-driven models and systems to optimize semiconductor design and analysis. Applies LLMs and deep learning to automate complex engineering workflows and improve efficiency in large-scale, data-intensive environments.

CORE COMPETENCIES & TECHNICAL SKILLS

AI / LLM / GenAI	LangChain, HuggingFace Transformers, OpenAI API, Llama, RAG Pipelines, Agentic Workflows, MCP, Prompt Engineering, Embedding models, Quantization
ML Frameworks	PyTorch, TensorFlow, Keras, Scikit-learn
ML Techniques	Supervised / Unsupervised / Semi-supervised Learning, ANN, CNN, GNN, LSTM, Transformer, Autoencoder, YOLO, VGG, ResNet, XGBoost, Random Forest, SVM
Data & Systems	Time-series Modeling, Feature Engineering, Statistical modeling, Outlier Detection, Dimensionality Reduction, Root cause analysis, FAISS, Qdrant
Languages & Tools	Python, R, MATLAB, SQL, Pandas, NumPy, FastAPI, CUDA, ReactJS, Node.js, Docker, Git, Jira
Semiconductor / EDA	Compact Modeling, Process Variation Analysis, TCAD, HSPICE, GAA/FinFET Device Simulation

PROFESSIONAL EXPERIENCE

Post-Doctoral Researcher/AI Engineer | Tzu Chi Hospital — Research Dept., Taipei, Taiwan *Dec 2025 – Present*

- Built an AI-powered Evidence-Based Medicine (EBM) platform that automates research retrieval, screening, and structured extraction (PICO, relevance scoring) using LLM-based RAG pipelines reducing manual review time by ~60%.
- Developed Transformer-based models for longitudinal EHR time-series analysis enabling early disease-risk prediction; engineered features across irregularly sampled, high-dimensional clinical datasets.
- Built an LLM-based academic progress assistant integrated with an institutional e-learning platform, providing students real-time, privacy-compliant visibility into course completion and graduation status.
- Resolved end-to-end system bottlenecks across data ingestion, SQL transactions, and model inference to improve pipeline robustness and throughput.

Post-Doctoral Researcher | Academia Sinica, Taipei, Taiwan *Feb 2025 – Nov 2025*

- Designed and automated end-to-end genomic data pipelines (QC → normalization → feature engineering → modeling), reducing processing time by ~50% and enabling reliable ML predictions under high-dimensional, low-sample constraints.
- Developed machine learning models (logistic regression, random forest, SVM) for cancer treatment response prediction from RNA-seq data, achieving AUC ~0.75.
- Deployed logistic regression models (glmnet) via an interactive R Shiny application, enabling user-driven inference on gene expression data, containerized with Docker for reproducible deployment.

EDUCATION

Ph.D., Electrical Engineering & Computer Science (EECS) | National Yang-Ming Chiao Tung University, Taiwan
Sept 2018 – Dec 2023

Research: Deep Learning for Emerging GAA MOSFET Device and circuit Modeling

- Conducted research and analyzed large complex non-linear dataset of semiconductor nano-devices such as Nanosheet, CFET to extract valuable insights and ensuring tape-out readiness.
- Developed ANN/LSTM predictive models for process variation effects (gate length, oxide thickness, work-function fluctuation) on device DC/AC/RF electrical characteristics, reducing TCAD simulation time by >40%.
- Applied CNN and dimensionality reduction (DR) to identify problematic OPC mask patterns, accelerating lithography correction workflows while maintaining prediction accuracy.
- Developed compact neural network models as drop-in replacements for TCAD/HSPICE simulators, co-designed hybrid TCAD+SPICE datasets for semi-supervised training to minimize labeled data requirements.
- Collaborated with cross-functional teams and published work in International Journals (IEEE-TED, IEEE Transactions on Microwave Theory & Techniques, Elsevier, IEEE Access) and conferences.

Master, Electronics & Communication Engineering | Jaypee Institute of Information Technology University, India
Aug 2010 – May 2012

- Dissertation: Image Enhancement using Thresholding & Histogram Equalization (HE), built full evaluation pipeline for brightness-preserving HE frameworks using MATLAB and Python.

RESEARCH PROJECTS

1. **ChimeraChat: RAG-Powered LLM Chatbot for ChimeraX:**
   - Built an LLM-driven chatbot that converts natural language into executable ChimeraX commands for molecular visualization and analysis.
   - Implemented a RAG pipeline using ChimeraX documentation to ground LLM outputs, improving accuracy and domain-specific command generation.
   - Integrated OpenAI API for inference and developed a web-based UI (HTML/CSS) to enable interactive, real-time user queries.
2. **LLM Multi-Task Content Generator** | Generative AI / NLP / Full-Stack AI Application:
   - Developed an on-premise AI platform leveraging LLMs to automate systematic reviews and meta-analysis, integrating multi-source literature retrieval with NLP-driven summarization, PICO extraction, and quality assessment.
   - Built a scalable, privacy-focused system with parallelized processing, structured inference pipelines, and an interactive dashboard for PRISMA reporting and efficient evidence synthesis.
3. **MERN-based-Shopping-Cart-Application** | Full-Stack Web Development / REST API Design:
   - Developed a full-stack MERN e-commerce application with RESTful APIs, enabling product browsing, cart management, and secure JWT-based user authentication.
   - Built a responsive React/Redux frontend with seamless Axios integration and a modular Node.js/Express backend using MongoDB for scalable data management.
4. **Hybrid Compact Model for Circuit Simulation:**
   - Neural network-based transistor compact model trained on hybrid TCAD/SPICE datasets with dynamic loss weighting, reduced TCAD simulation requirements by >50%.

PUBLICATIONS

Journals:

1. **R. Butola** et al. "Compact Model Build upon Dynamic Weighting Artificial Neural Network Approach for Complementary Field Effect Transistors," IEEE Transactions on Electronic Devices (IEEE-TED), vol. 71, Jan. 2024.
2. S. R. Kola, Y. Li and **R. Butola**, "Statistical Device Simulation and Machine Learning of Process Variation Effects of Vertically Stacked Gate-All-Around Si Nanosheet CFETs," in IEEE Transactions on Nanotechnology, vol. 23, April, 2024.
3. **R. Butola** et al. "Application of Long Short-Term Memory Modeling Technique to Predict Process Variation Effects of Stacked Gate-All-Around Si Nanosheet Complementary-Field Effect Transistors," Computers & Electrical Engineering, Elsevier, vol. 105, Jan. 2023.
4. **R. Butola** et al. "Artificial Neural Network-Based Modeling for Estimating the Effects of Various Random Fluctuations on DC/Analog/RF Characteristics of GAA Si Nanosheet FETs," IEEE Trans. Microwave Theory & Techniques, vol. 70, no. 11, Nov. 2022.
5. **R. Butola** et al. "A Machine Learning Approach to Modeling Intrinsic Parameter Fluctuation of GAA Si Nanosheet MOSFETs," IEEE Access, vol. 10, July 2022.
6. **R. Butola** et al. "A Comprehensive Technique based on Machine Learning for Device and Circuit Modeling of Gate-All-Around Nanosheet Transistors," IEEE OJ-Nano, vol. 4, Oct. 2023.
7. P. Parashar, C. Akbar, T. S. Rawat, S. Pratik, **R. Butola**, S. H. Chen, Y-S Chang, S. Nuannimnoi, and A. S. Lin "Intelligent Photolithography Corrections Using Dimensionality Reductions," IEEE Photonics Journal, vol. 11, no. 5, Oct. 2019.
8. P. Parashar, C. H. Chen, C. Akbar, S. M. Fu, T. S. Rawat, S. Pratik, **R. Butola**, S. H. Chen, A. S. Lin. "Analytics-statistics mixed training and its fitness to semi-supervised manufacturing" PLoS ONE Journal, 2019.

Conferences:

1. **R. Butola** et al. "A DNN-Based Compact Modeling Technique for GAA Si NS FETs and Its Application in CMOS Circuit Simulation," International conferences on the science and technology of devices and materials (SSDM), Japan, 2023.
2. **R. Butola** et al. "Estimating the Process Variation Effects of Stacked Gate All Around Si Nanosheet CFETs Using Artificial Neural Network Modeling Framework," IEEE International Conference on Nanotechnology (IEEE-NANO), Spain, 2022.
3. **R. Butola** et al. "Deep Learning Approach to Estimating Work Function Fluctuation of Gate-All-Around Silicon Nanosheet MOSFETs with A Ferroelectric HZO Layer," IEEE-EDTM, Japan, 2022.
4. **R. Butola** et al. "Machine Learning Approach to Characteristic Fluctuation of Bulk FinFETs Induced by Random Interface Traps," IEEE-ISQED, USA, 2022.
5. **R. Butola** et al. "Deep Learning Approach to Modeling and Exploring Random Sources of Gate-All-Around Silicon Nanosheet MOSFETs," International Symposium on VLSI Technology, Systems and Applications (VLSI-TSA), Hsinchu, Taiwan, 2022.